

[6584]-218

B.E. (Mechanical)

DYNAMICS OF MACHINERY

(2019 Pattern) (Semester - VII) (402042)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data wherever necessary.

- Q1) a)** For the spring mass system shown in Fig. 1a),
If $K_1 = 2 \text{ N/mm}$, $K_2 = 1.5 \text{ N/mm}$, $K_3 = 3 \text{ N/mm}$, $K_4 = K_5 = 1.5 \text{ N/mm}$,
Determine mass 'm' if the system has a natural frequency of 12 Hz. [9]

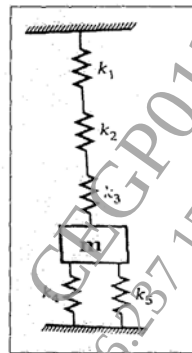


Fig. 1a)

- b) Enlist elements of a vibratory system and explain with neat diagram mathematical modeling of a bicycle. [8]
- OR
- Q2) a)** In a Single Degree of Freedom vibratory system, a suspended mass of 16 kg makes 30 oscillations in 22 seconds. The amplitude of oscillations decreases to 0.25 of initial value after 5 oscillations. [9]
Determine the
- i) Stiffness of the spring
 - ii) Logarithmic decrement
 - iii) Damping factor
 - iv) Damping Coefficient
- b) Discuss importance of damping in vibratory system. Explain various categories of damping based on damping ratio (ξ). [8]

P.T.O.

- Q3) a)** An electric motor running at 500 rpm is supported on a spring and a dashpot. The spring has the stiffness of 6400 N/m and dashpot offers a resistance of 400 N at 4.0 m/sec. The unbalanced mass of 0.6 kg rotates at 5 cm radius and the total mass of vibratory body is 16 kg. **[10]**

Calculate:

- Damping factor
 - Amplitude of vibration and phase angle
 - Resonant speed and resonant amplitude
 - Force exerted by spring and dashpot on motor
 - Resultant force
- b) Explain in detail with neat sketch forced vibration due to rotating unbalance. **[8]**

OR

- Q4) a)** A machine having mass of 200 kg is mounted on spring and damper. The total stiffness of spring is 7.84×10^5 N/m while damping ratio of damper is 0.15. A vertical harmonic force $F = 392 \sin (314.15t)$ N acts on the machine. For the steady state vibrations of the system, **[10]**

Calculate,

- amplitude of vibration
 - the transmissibility
 - transmitted Force
 - phase difference between transmitted force and excited force
- b) Explain quality factor and half power bandwidth. **[8]**

- Q5) a)** For the vibratory system shown in Fig. 5 a), determine two natural frequencies and corresponding modes of vibration. **[10]**

Given, $m_1 = 2$ kg, $m_2 = 3$ kg
 $K_1 = K_2 = K = 30$ N/m

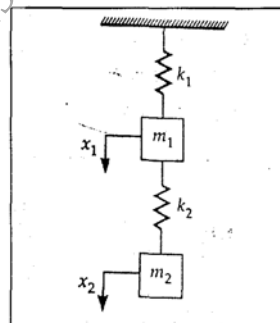


Fig. 5 a)

- b) Explain the concept of torsionally equivalent shaft and derive equation for its equivalent length. **[7]**

OR

- Q6) a)** For the Spring mass system shown in Fig. 6 a), find Eigen values and Eigen vectors by matrix method [10]

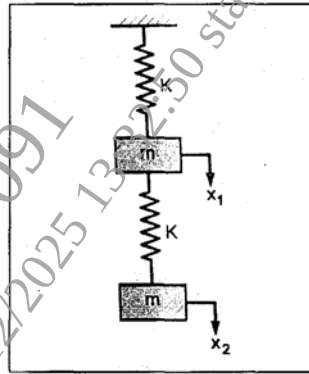


Fig. 6 a)

- b) Explain the torsional vibrations of geared system neglecting inertia of gears. [7]
- Q7) a)** Explain time domain and frequency domain techniques for vibration monitoring. [10]
- b) List the vibration measuring instruments. Explain any one in detail. [8]

OR

- Q8) a)** Explain in detail various types of sound field in the vicinity of a sound source. [10]
- b) What is the need of vibration control? Explain any two methods of vibration control. [8]

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